Evaluating Deep Reinforcement Learning Policies for LiDAR-Based Navigation: From Sensor Degradation to Complex Environments

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***Abstract*—Deep Reinforcement Learning (DRL) has emerged as a powerful approach for autonomous mobile robot navigation. However, transferring policies from pristine simulated environments to the real world remains a challenge due to the sim-to-real gap, which is heavily influenced by sensor noise and resolution constraints. This paper presents a systematic, three-phase study evaluating the resilience and spatial generalization of DRL agents utilizing only LiDAR data for hallway navigation. First, we establish boundaries for realistic sensor noise injection, identifying thresholds where training collapses. Second, we analyze the impact of sensory resolution by training and cross-testing agents configured with 360 and 90 LiDAR rays under diverse operational constraints. Finally, we benchmark different DRL architectures in a complex hallway layout, adapted manually from the original floor plan. Our results highlight that while continuous PPO agents exhibit robust zero-shot resilience to sensor degradation, they struggle with spatial generalization when transferred to novel layouts. In contrast, discrete DQN agents, when trained within the complex environment, achieve superior performance, reaching a 65% success rate compared to PPO's 46% post-fine-tuning performance. Finally, we conduct a preliminary assessment of the PPO architecture's reactivity to dynamic obstacles, identifying significant challenges in temporal awareness that warrant further investigation.**

***Index Terms*—Deep Reinforcement Learning, Autonomous Navigation, Sim-to-Real Gap, LiDAR Resolution, Spatial Generalization, PPO vs DQN.**

I. Introduction

Autonomous local navigation in indoor environments, such as narrow hallways and corridors, represents a classic yet critical problem in mobile robotics. To circumvent the extensive hand-tuning required by traditional analytical methods, Deep Reinforcement Learning (DRL) has been widely adopted. DRL allows an agent to map high-dimensional sensory inputs, like LiDAR range graphs, directly into continuous or discrete motion commands by maximizing a cumulative scalar reward function.

Despite showing high success rates in idealized simulation environments, deploying these learned policies onto physical robotic platforms often exhibits severe performance drop-offs, a phenomenon known as the 'sim-to-real gap.' This issue stems largely from overfitting to pristine simulated data. Physical sensors are inherently subject to hardware limitations, thermal noise, and varying angular resolutions; if an agent is not exposed to these imperfections during its training phase, its policy becomes overly specialized to the noise-free metrics of the simulator, leading to catastrophic collisions when exposed to even minor real-world variations.

To tackle this challenge, this work presents a systematic study divided into three incremental phases. In Phase 1, we investigate the limit of noise endurance during training, demonstrating how hyperparameter noise levels can cause policy divergence. In Phase 2, we introduce a resolution sensitivity analysis, training and cross-testing agents with maximum (360) and sparse (90) LiDAR rays to determine the most computationally efficient and robust sensory setup. Lastly, in Phase 3, we analyze spatial generalization and algorithm performance by benchmarking Proximal Policy Optimization (PPO) against Deep Q-Networks (DQN) within a newly developed, complex hallway environment. This phase further explores the agents' robustness through an additional, targeted assessment of reactivity to dynamic obstacles, introducing a moving pillar to evaluate temporal perception beyond static navigation.

The primary contributions of this paper are twofold: (1) a comprehensive evaluation of how sensor noise and LiDAR ray density affect DRL policy robustness, and (2) a comparative analysis of DRL policy performance across complex structural variations, complemented by a preliminary assessment of reactivity to dynamic obstacles using continuous control algorithms.

The remainder of this paper is structured as follows. Section II presents related work in DRL autonomous navigation. Section III details our methodological approach, outlining the three experimental phases and algorithm configurations. Section IV discusses the experimental evaluation, including noise resilience, spatial generalization, and dynamic obstacle avoidance. Finally, Section V presents our conclusions and directions for future work.

II. Related Work

The application of DRL to autonomous navigation has been extensively studied using algorithms such as Deep Q-Networks (DQN) for discrete action spaces and Proximal Policy Optimization (PPO) for continuous control. Early implementations demonstrated that agents could learn effective obstacle avoidance behaviors solely from sparse distance measurements [6]. Standardized benchmarking pipelines for these implementations have been heavily accelerated by framework integrations like OpenAI Gym [2], which decouple the physical mechanics of simulation environments from the step-by-step state-action interfaces required by training agents.

Building upon these standardized architectures, libraries like Stable-Baselines3 [3] have provided highly optimized, reliable open-source implementations of state-of-the-art DRL algorithms. In particular, Proximal Policy Optimization (PPO) [4] has become a foundational paradigm for mobile robotics due to its clipped surrogate objective function, which prevents catastrophically large policy updates during training, thus ensuring stable convergence trajectories in continuous operational spaces. Conversely, Deep Q-Networks (DQN) [5] leverage value-based architectures combined with experience replay mechanisms to optimize discrete decision-making, providing a compelling benchmark for evaluating whether smooth continuous velocity commands outperform discrete turning intervals in structured layouts.

However, recent literature heavily focuses on mitigating the sim-to-real gap. Domain Randomization (DR) is a popular technique where sensor properties, friction coefficients, and environmental layouts are dynamically altered during training to force the neural network to learn a generalized policy. While adding Gaussian noise to simulated sensors is a standard practice, few works explore the exact thresholds where noise ceases to act as a regularizer and begins to induce training instability.

Furthermore, the precise trade-off between sensory resolution and operational resilience remains an active debate. Recent studies, such as the work by Olayemi et al. [7], evaluate the impact of LiDAR configurations on goal-based navigation, demonstrating that optimally configured fields of view (FOV) and lower-resolution beam arrays can still maintain high success rates, thereby reducing computational complexity and hardware costs. Conversely, although sparse configurations reduce processing bottlenecks, they risk missing fine structural details like sharp corners if not properly optimized.

Spatial generalization across unmapped topologies also presents a critical performance gap. As Packer et al. [8] emphasize, DRL algorithms are highly sensitive to subtle environmental changes and are prone to severe overfitting to the training distribution. Agents trained in uniform environments often struggle with extrapolation, failing when encountering novel architectural variations—such as bottleneck constrictions or L-junctions—due to the policy's inability to adapt beyond the specific conditions experienced during training. This paper aims to address these open questions by systematically mapping sensory performance boundaries using PPO, and spatial performance boundaries through a comparative analysis of PPO and DQN architectures.

III. Methodology

Our experimental architecture is implemented using the Webots simulator, utilizing differential drive robotic agents communicating with a Python-based server via ZeroMQ (ZMQ) sockets. The base communication and simulation structure builds upon the `drl-robot-nav-hallways` framework [1], which we adapted to implement our specific PPO and DQN navigation paradigms. This decoupling allows for headless training acceleration while preserving full physics interactions.

A. Phase 1: Noise Threshold Determination

To evaluate the impact of environmental stochasticity during training, we established two baseline paradigms using the standard continuous PPO architecture. The initial training environment consisted of a three-corridor layout: one strictly linear corridor, and two featuring asymmetrical wall protrusions that forced the agents to execute minor path deviations. The models evaluated were:

- **Model A (Ideal Baseline):** Trained under perfect simulated conditions with 360 LiDAR rays and zero added sensor noise.
- **Model B (Over-Noised Control):** Trained with an extreme Gaussian noise factor of $\sigma = 0.5$ injected into the normalized range vectors.

As detailed in the results section, Model B suffered from complete training collapse due to severe state-space obfuscation, leading us to establish an optimal, realistic training noise threshold of $\sigma = 0.05$ for all subsequent experiments.

B. Phase 2: Resolution and Ray-Density Analysis

Using the optimized noise factor ($\sigma = 0.05$), Phase 2 explores sensory density by adjusting the number of range beams fed into a Convolutional Neural Network (CNN) feature extractor. Two distinct PPO models were trained for 1,000,000 timesteps on the initial environment:

- **Model C (High Resolution):** Configured with a 360-ray LiDAR array, representing maximum structural visibility.
- **Model D (Low Resolution):** Configured with a 90-ray LiDAR array, representing cost-effective, low-end physical sensors.

Both models were subjected to cross-evaluation across diverse noise profiles. Although Model D demonstrated high theoretical efficiency, underlying framework version constraints during deployment led us to select Model C as the foundational baseline for Phase 3 spatial generalization testing.

C. Phase 3: Spatial Generalization and Algorithmic Benchmarking

To test spatial generalization, we initially invested significant effort into designing a custom map from scratch, utilizing a parsing script to programmatically generate the 3D topology. However, evaluating Model C in this novel environment resulted in catastrophic policy failure. The agents exhibited severe oscillatory behaviors (trembling) and remained completely static, failing to initiate any navigation sequence. Consequently, this custom map was discarded.

Instead, to robustly evaluate the agent's adaptability, we opted to systematically increase the complexity of the original environment through manual modifications within the Webots interface. The agents were relocated into three distinct, previously unused corridors, each presenting a unique architectural challenge:

- **Robot 0:** Positioned in a wide corridor requiring a challenging 180-degree U-turn, followed by the navigation around a static obstacle placed at the exit of the turn.
- **Robot 1:** Assigned to a pathway characterized by a 90-degree turn, where structural wall segments create a funneling effect that directs the trajectory towards a static pillar positioned at the inner vertex of the corner, demanding precise obstacle avoidance during the transition.
- **Robot 2:** Placed in a highly restrictive corridor featuring a severe bottleneck constriction and a central obstacle.

Static obstacles were manually introduced to these new paths, establishing the new environment. All Phase 3 spatial generalization tests, fine-tuning, and DQN benchmarking were conducted exclusively within these newly assigned areas to properly evaluate adaptability to unmapped topologies.

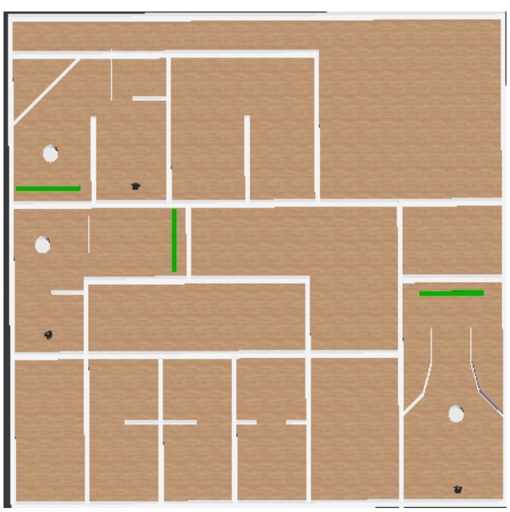

Fig. 1. Complex layout utilized for Phase 3 evaluation. This environment was adapted from the original map by relocating agents to unmapped corridors and introducing static obstacles to benchmark spatial generalization.

The evaluation in this complex layout was conducted in three steps:

- **Fine-Tuning (PPO):** Model C, previously trained for 1,000,000 timesteps, underwent additional fine-tuning for 1,000,000 timesteps, totaling 2,000,000 cumulative timesteps. During this process, we observed significant volatility in the training metrics, with the success rate peaking at 0.76 before stabilizing at a final value of 0.56.
- **From-Scratch Training (DQN):** The DQN model was trained entirely from scratch on the new map for 800,000 timesteps. This agent demonstrated a more stable convergence, reaching a peak success rate of 0.57 and concluding the training phase with a consistent 0.53.
- **Dynamic Obstacle Testing:** To evaluate temporal reactivity, a moving pillar was introduced into one of the corridors.

Within this new environment, we benchmark two foundational DRL paradigms—continuous PPO and discrete DQN—utilizing identical CNN architectures to process the optimal ray configuration determined in Phase 2, thereby isolating the impact of the control strategy on navigation performance.

D. Evaluation Metrics

Performance is quantified over a fixed evaluation budget of 60,000 environment steps per configuration, which yielded an average of approximately 100 episodes per agent. The primary metric is defined as:

- **Success Rate (%):** The percentage of episodes where the agent successfully reaches the designated goal zone without triggering a Bumper collision.

Beyond the success rate, we perform a qualitative analysis of agent behaviors, including navigation patterns, oscillatory movements, or failure modes observed through direct manual inspection during the simulation runs. These observations provide context to the quantitative results, particularly when explaining the specific causes of navigation failure or sub-optimal agent performance in complex topologies.

IV. Results and Discussion

A. Phase 1: Noise Injection and Policy Collapse

The initial phase of our study established performance benchmarks for the ideal baseline (**Model A**, trained with 360 rays and $\sigma = 0.0$) and evaluated the consequences of extreme training stochasticity (**Model B**, trained with 36 rays and $\sigma = 0.5$). After a training budget of 1,000,000 timesteps, both models were subjected to cross-testing environments to isolate specific vulnerabilities.

1) Resilience to Sensory Resolution Reduction: As summarized in Table I, Model A exhibited remarkable resilience when its active LiDAR beam density was severely truncated during test episodes, maintaining a flawless 100% success rate down to 90 rays, and only declining to 98.63% with just 36 rays. This suggests that the spatial representation learned by Model A's CNN feature extractor does not rely on high-frequency angular sampling, but rather on generalized geometric boundaries.

Conversely, Model B failed completely (0% success rate) across all resolutions. Since Model B was explicitly trained with a sparse setting of 36 rays, its failure at 36 rays during testing proves that its weakness was not caused by LiDAR density itself, but by a foundational convergence failure during training.

TABLE I
PHASE 1 SUCCESS RATE (%) UNDER REDUCED RAYS

Model	**360 Rays**	**180 Rays**	**90 Rays**	**36 Rays**
Model A	100.00%	100.00%	100.00%	98.63%
Model B	0.00%	0.00%	0.00%	0.00%

2) Vulnerability to Gaussian Noise Injection: The effects of sensor noise injection posed a significantly greater challenge than ray-density reduction, as detailed in Table II. While Model A preserved its perfect 100% success under a mild noise variance of $\sigma = 0.1$, its performance suffered a catastrophic drop-off when the noise factor was raised to $\sigma = 0.2$ (26.6%) and $\sigma = 0.5$ (28.2%). Model B failed across all noise test-beds, showing that training under severe noise ($\sigma = 0.5$) did not produce the expected domain robustness.

TABLE II
PHASE 1 SUCCESS RATE (%) UNDER NOISE INJECTION

Model	**Noise** $\sigma = 0.1$	**Noise** $\sigma = 0.2$	**Noise** $\sigma = 0.5$
Model A	100.00%	26.60%	28.20%
Model B	0.00%	0.00%	0.00%

This empirical breakdown highlights that high-variance noise ($\sigma \geq 0.2$) completely obfuscates close-range wall boundaries, preventing the agent from computing stable centering gradients. For Model B, an extreme training hyperparameter choice like $\sigma = 0.5$ injected too much entropy into the state-space, preventing the network from establishing a basic correlation between linear velocity and safe spatial margins. Consequently, an operational threshold of $\sigma = 0.05$ was mandated for Phase 2 and Phase 3.

B. Phase 2: LiDAR Resolution Cross-Testing

Table III outlines the cross-evaluation matrix for Model C and Model D across various ray restrictions.

TABLE III
PHASE 2: CROSS-RESOLUTION EVALUATION MATRIX

Model (Train Setup)	**Test Rays**	**Success Rate (%)**
Model C (360 Rays)	360	100.00%
Model C (360 Rays)	180	100.00%
Model C (360 Rays)	90	100.00%
Model C (360 Rays)	36	100.00%
Model D (90 Rays)	360	100.00%
Model D (90 Rays)	180	100.00%
Model D (90 Rays)	90	100.00%
Model D (90 Rays)	36	100.00%

Discussion on Resolution: The cross-evaluation experiment yielded a uniform 100% success rate across all structural testing configurations for both Model C and Model D. This absolute resilience indicates that the continuous control policies managed to abstract the generalized boundary limits of the corridor layout rather than over-fitting to specific ray densities.

Crucially, the performance of Model D proves that downscaling the sensory training input to 90 rays does not degrade the agent's spatial awareness. Thanks to the mathematical interpolation layer embedded within the environment architecture, Model D remains completely immune to abrupt hardware resolution degradation, maintaining flawless navigation safety even when downgraded to a highly sparse array of 36 rays. From an engineering standpoint, this highlights a significant optimization opportunity: autonomous wheelchairs can achieve maximum operational safety utilizing lower-resolution, cost-effective LiDAR sensors, successfully reducing both computational overhead and hardware deployment budgets without compromising safety.

C. Phase 3: Spatial Generalization (PPO vs DQN)

The results of the algorithmic benchmarking within the programmatically generated complex map are summarized in Table IV.

TABLE IV
PHASE 3: ALGORITHM PERFORMANCE IN COMPLEX LAYOUT

Algorithm	**Evaluation Condition**	**Success Rate (%)**
PPO (Continuous)	Zero-Shot Transfer	8.24%
PPO (Continuous)	Post Fine-Tuning	46.00%
DQN (Discrete)	Trained From Scratch	65.00%

When transferring the Phase 2 continuous model (Model C) to the `smart-wheelchairs-complex.wbt` map without prior exposure, the policy struggled significantly with the novel topologies, yielding an 8.24% overall success rate. Specifically, Robot 0 achieved a 43.75% success rate in the sweeping turn, while Robot 1 (pillar) and Robot 2 (bottleneck) completely failed (0%). This highlights the severe spatial overfitting of the initial policy trained on simpler corridors.

Subsequent fine-tuning of this PPO model for 2,000,000 timesteps improved the overall success rate to 46% (Robot 0: 0%, Robot 1: 100%, Robot 2: 47.37%). As visually demonstrated in Fig. 2, Robot 1 successfully negotiated its assigned pathway, navigating the 90-degree turn and the associated static obstacle. In contrast, Robot 0 completely lost its previous ability to navigate the 180-degree turn, while Robot 2 struggled to traverse the bottleneck constraints.

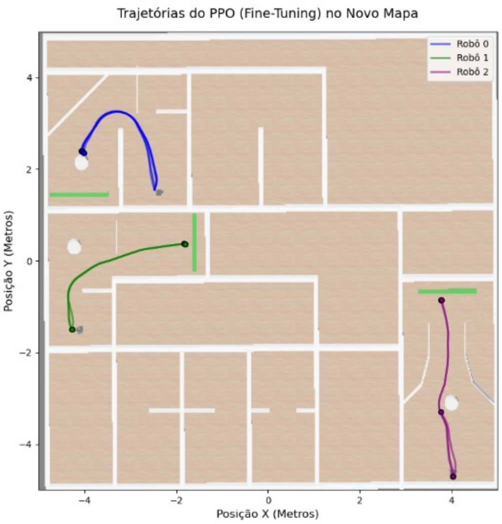

Fig. 2. Recorded trajectories of the three robots utilizing the fine-tuned PPO model in the complex environment. Robot 1 successfully navigates the corridor, while Robots 0 and 2 exhibit early termination and sub-optimal routing.

In contrast, the DQN model, trained from scratch for only 800,000 timesteps, outperformed the PPO architecture, achieving a 65% overall success rate. While Robot 0 still failed the wide turn (0%), the DQN agent demonstrated exceptional spatial generalization in the more constrained environments, achieving 100% success rates for both Robot 1 (pillar) and Robot 2 (bottleneck). This superior performance was heavily dependent on the action-selection policy. When operating under a deterministic policy (deterministic=True), Robots 0 and 2 stopped prematurely, evidently learning a sub-optimal, risk-averse behavior to avoid bumper penalties. Enabling a stochastic policy (deterministic=False) effectively resolved the hesitation observed under deterministic conditions. However, we noted occasional regressive behavior near the goal line, particularly with Robot 2. This suggests that the agent's learned value function exhibits high sensitivity to proximity penalties in the terminal phase of navigation; the agent occasionally prioritizes bumper-collision avoidance over immediate goal acquisition, indicating a clear trade-off in the implemented reward shaping process.

D. Robustness to Dynamic Obstacles

To evaluate reactive navigation, a separate environment variant (smart-wheelchair-complex-mov.wbt) was developed. A moving pillar was introduced into Robot 1's trajectory using the fine-tuned PPO model. The agent completely failed to account for the dynamic obstacle, maintaining its predefined path without any reactive adjustment; consequently, whenever the pillar's trajectory intersected with the agent's path, a collision occurred. This indicates that while the LiDAR-only CNN architecture effectively maps static geometric boundaries, it lacks the temporal awareness required to extrapolate the velocity vectors of moving objects, treating them instead as non-existent or static background features.

V. Conclusion and Future Work

This paper presented a rigorous analysis of LiDAR-only autonomous navigation via DRL under sensory and spatial constraints. Our investigations concluded that while sensor noise ($\sigma = 0.05$) can regularize training, extreme variances prompt mathematical divergence. Furthermore, our cross-resolution study demonstrates that lower-resolution LiDAR setups (down to 90 rays) can maintain flawless navigation safety when properly optimized.

Finally, our benchmark in complex environments highlights critical performance gaps between continuous and discrete control architectures. The discrete DQN algorithm demonstrated superior spatial adaptability (65% success rate) compared to the fine-tuned PPO model (46%). However, the preliminary assessment of the PPO architecture revealed significant challenges in reactivity to dynamic obstacles, highlighting a limitation in its instantaneous LiDAR mapping approach. Future work will focus on integrating recurrent neural network (RNN) layers or frame-stacking techniques to provide agents with the temporal awareness necessary for dynamic obstacle avoidance in unpredictable real-world environments.

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